Overlay networks

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Outline

1 Subject Information

Professors and rooms

- Óscar Esparza
  - Office: C3-310
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- Lectures: Wednesdays 12:00-14:00 in C3002.
- Lab sessions: Fridays 11:00-13:00 in C3-S102:
  - Group 41: "A" weeks
  - Group 43: "B" weeks

Course contents

1. Introduction: concept of overlay
2. In lectures you will be introduced about:
   1. Introduction to the topics of the lab sessions (IP tunnels, IP Multicast and SIP)
   2. P2P (peer-to-peer) networks.
3. In lab sessions you will test:
   1. Tools to construct overlays: IP tunneling, IP multicast.
   2. An example of multimedia overlay: SIP.

There will be material (slides, papers, etc.) of all the topics available in Atenea.

Lab sessions

1. Virtual machine to download (OVA file).
   http://bohr.upc.es:8000/Networks/ova/
2. Access simtools via browser (ravada).
   https://rvd4.upc.edu/
   Virtual machine: TCGI-debian-20-21-1Q
3. Previous concepts you must know:
   1. Basics of linux operating system and networking.
   2. IP: addresses, subnetting, static routing, header fields, MTU, etc.
   3. UDP: ports, header fields.
   4. TCP: ports, connection establishment and ending, header fields, MSS, etc.
   5. iptables and firewall configuration.

Evaluation

There is no final exam. All the assessment is continuous:

- 55%: Theoretical assessment (quizzes):
  - IP tunnels, p2p, IP multicast and SIP.
  - The overall grade of the quizzes will be weighted according to the number of questions of each quiz.
  - The prospective dates are in the schedule of the subject, in Atenea.
- 35%: Lab exam:
  - Wednesday 24th of May.
  - Individual and practical (with the lab tools).
  - In the A2-S10X rooms.
- 10%: Attendance.